

Appendix E

Crop Characteristics

Table E.1: Defining characteristics of crops

	Horse Bean	Einkorn	Emmer	Lentils	Peas	Olives	Grapes	Grass Peas	Bitter Vetch	Barley	Free Hull Barley
Yield	500	500	500	300	150	600	600	500	400	500	500
Units	Ha	Ha	Ha	Ha	Ha	Ha	Ha	Ha	Ha	Ha	Ha
Mcal kg ⁻¹	3	3	3	3	3.25	4	3	3	3	3	3
Fallow Ratio	1	1	1	1	1	0	0	1	1	1	1
Fallow MCals	500	500	500	500	500	500	500	500	500	500	500
Fodder Kg	300	300	300	300	100	0	0	300	300	300	300
Fodder MCals	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8	1.42	0.8

The values used in Table E.1 have been derived from a variety of sources. It can rightfully be argued that any one of these values might have been higher or lower; for example there is no possible way to know the exact yield of any crop in the past, so how can a yield of 500 kg/ha be justified? Even today, farmers do not know what the yield of any particular field will be until they harvest it, and that yield can be very different even from another field on ‘the other side of the fence’. The values used are, for the purposes of the model, approximations that, even if not perfect, will not invalidate the outcome of the model given the imprecise nature of many of the other variables which are used (e.g. variables where minimum and maximum values have been adopted, such as population levels, will ‘absorb’ imprecisions regarding the input factors used to define crops). Additionally, there are other major influencing factors that cannot be known such as sowing rates, harvesting efficiency, and so on.

In the interests of consistency for use in the model, and to compensate for loss of energy due to cooking and other factors (as well as to simplify the model), I have opted to use conservative and ‘even’ values which have been based on a large number of sources ranging from personal communications and animal science websites to aca-

demic journals and PhD theses. The final numbers, therefore, may not perfectly match any one source in their entirety, but they are reasonable approximations. Some of the primary references used are as follows. Yield values have been adapted from Campbell (2009, Tables 5.3 and 5.4) and (Ellison 1981, 40). The food value for the crops have been based upon USDA figures, which, for example, state that wheat has approximately 3.2 to 3.4 Mcals per kg and that pulses contain 3.2 to 3.5 Mcals (USDA 2009). Straw and chaff (fodder) values adapted from Simms and Russell (1997, Table I).

Appendix F

Animal Characteristics

Table F.1: Defining characteristics of animals

	Sheep	Pig	Cow	Goat
Mcalories Usable	3.200	3.200	3.200	3.200
Average Kill Weight	50.00%	50.00%	50.00%	50.00%
Weaning Weight	30	100	400	30
Adult Weight	8	2	50	8
Grow Time	60	200	625	60
Death Rate	40	40	50	40
Conception Efficiency	30.00%	30.00%	20.00%	30.00%
Sex Maturity	80.00%	70.00%	85.00%	80.00%
Breeding Years	18.00	18	18	18
Babies Birth ⁻¹	5	5	8	5
Wean Age	1	6	1	1
Gestation	60	30	60	60
Oestrous Cycle	150	115	279	150
Lactation Time	21	21	21	21
Ratio F:M breedstock	210	30	240	210
	20	20	20	20
Desired Herd Size	50	0	10	50
Mixed Herd Numbers	25	0	2	25
Milk grams day ⁻¹	0.94	0	6	0.9
Milk Mcals Kg ⁻¹	0.940	0	0.94	0.940
Gestation Req'd	3.970	3.970	6	3.970
Lactation Req'd	5.410	5.410	7.5	5.410
Maintenance Req'd	3.200	0	5.8	3.2
Juveniles Req kg ⁻¹	0.030	0.030	0.030	0.030
Land Energy Hectare ⁻¹	1500	1500	1500	1500
Fodder Fed From Crop	Crop10	Crop10	Crop10	Crop10
	Barley	Barley	Barley	Barley
Grams ⁻¹ Straw/Chaff	1000	0	4956	1000
Grams ⁻¹ Grain	750	750	1500	750
Days on supplemental feed	90	90	90	90

The values used in Table F.1 have, like the crop species characteristics (Appendix E

on page 583) been derived from a variety of sources (e.g. Bhattacharyal and Harb (1973, Tables 2 and 4), Danti (2000, 57-64), Breman and de Wit (1983, 1344-1346) Balasse and Tresset (2002, 858) and Tennessee Health and Science Center (2009)). It can again rightfully be argued that many of these values could have been higher or lower; for example there is no possible way to know what the precise weaning age was, or precisely how many males were kept in relation to females and so on. The purpose of the large number of input variables used to model the herds is to allow for a consistent and systematic method for any species to be defined for use in the model. Slight inaccuracies of these variables have very little effect on the overall outcome of the model. Conceivably, the factors that would tend to have the most effect tend to be values that remain relatively consistent with any specific species (such as: offspring per birthing event, gestation time, age of sexual maturity) and are thus less likely to be significantly inaccurate.